EU Regulatory Power and Market Credibility: Evidence from the Green Bond Market

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Abstract

Corporate greenwashing poses a significant challenge to the integrity of environmental finance, as mobilizing funds to meet the Paris Agreement goals requires substantive sustainability efforts. This article examines the introduction of the European Union's Green Bond Standard (EU GBS) to answer why and how public authority matters in markets previously governed by private standards. I argue that in markets characterized by fragmented private governance and persistent information asymmetries, EU regulatory power can alter market credibility by reordering reputational incentives. Rather than merely imposing compliance costs, the introduction of a credible EU standard increases short-term market uncertainty, prompting investors to reassess the credibility of existing green financial instruments. Using a regression discontinuity in time design, I find that green bond yields increased significantly in the immediate aftermath of the legislation. Moreover, the effect varies, with firms boasting stronger environmental reputations experiencing smaller increases. This heterogeneity suggests that investors rely on reputational heuristics to navigate regulatory transitions, disproportionately penalizing perceived environmental laggards. Ultimately, this research demonstrates the EU's climate leadership in promoting stricter oversight to curb corporate greenwashing, encouraging more informed investment decisions, and driving the transition to a greener economy.

Keywords: European Union, Sustainable Finance, Regulatory Governance, Environmental Governance, Comparative Political Economy

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1 Introduction

The emergence of green bonds as a financial instrument designed to fund environmentally sustainable projects marked a significant shift in global capital markets. Over the past decade, the green bond market has experienced exponential growth, driven by increasing investor demand for sustainable finance and the growing urgency of addressing climate change. However, this rapid expansion has also exposed the market to critical challenges, most notably the issue of greenwashing. The absence of standards that are universally accepted and enforced has led to a heterogeneous market creating information asymmetry that can undermine investor confidence.

Amidst this backdrop, the European Union (EU) introduced its Green Bond Standard (EU GBS), a regulatory framework aimed at enhancing transparency, credibility, and comparability in the green bond market. The EU GBS sets out stringent criteria for what qualifies as a green bond, including alignment with the EU Sustainability Taxonomy, mandatory reporting, and external verification. As the EU is a significant player in global finance and a leader in climate policy, the introduction of this standard is poised to have a substantial impact on the global green bond market.

This paper explores the impact of the EU GBS on green bond pricing dynamics, particularly focusing on how the new standard influences bond yields—a key indicator of investor confidence and market conditions. Specifically, the paper hypothesizes that the EU GBS will lead to an overall increase in green bond yields as investors become more cautious in the short-term and therefore ask for a higher return of their investment in the face of the uncertainty that the EU standard brings, reflecting heightened scrutiny over the environmental credentials of issuers.

To test these hypotheses, the paper employs a Regression Discontinuity in Time (RDiT) design, leveraging the EU GBS's announcement as a natural experiment. The analysis also investigates the heterogeneity of these effects, focusing on bonds with and without external verification, as well as issuers classified as environmental leaders or laggards based on ESG scores and controversy indicators. My findings suggest that investors are applying a "credibility premium". Where a green bond issuer has stronger environmental credentials, there is little market adjustment following the introduction of the EU's more stringent criteria for certification. Where a green bond issuer has weaker environmental credentials, a credibility gap premium is applied to cover for the danger of holding an asset that may fall foul of the new public regulation in the short term.

This article makes three broader contributions to comparative European political economy. First, it advances a general theoretical argument about the conditions under which public authority reshapes market behavior in policy domains characterized by information asymmetry and extensive private governance In particular, it speaks to the work that seeks to explain the interaction between politico-economic effects of environmental regulation and the firm-level characteristics that shape the heterogeneity of these effects (Bayer 2023; Bechtel, Genovese, and Scheve 2019; Brulle 2021; Cheon and Urpelainen 2013; Distelhorst and Locke 2018; Genovese and Tvinnereim 2019; Kelsey 2018; Kennard 2020; Kim, Urpelainen, and Yang 2016; Lerner and Osgood 2023; Meckling 2015; Monogan III, Konisky, and Woods 2017). By providing empirical evidence on the impact of the EU GBS, this paper not only sheds light on the immediate consequences for green bond yields but also offers insights into the broader implications for market behavior and corporate environmental practices.

Secondly, it engages, more broadly, with the literature in environmental politics that addresses EU environmental leadership (Oberthür and Dupont 2021; Schulze and Tosun 2013; Smeets and Beach 2023; Vogel 2003) by showing how the credibility of the EU as a climate leader can be used as a signalling tool for market participants to avoid reputational risks. Particularly, while there is robust evidence within this literature regarding the long-term impacts of EU policies, evidence that examines the short-term effects of such policies is rare

(Bechtel and Schneider 2010). I contribute to debates on EU regulatory power by identifying the political and economic consequences of EU standard-setting beyond formal compliance, showing how EU authority can reshape expectations and behavior in transnational markets. Together, these findings speak to broader comparative questions about public–private governance, regulatory legitimacy, and the role of supranational institutions in structuring market outcomes. Ultimately, I underscore the critical role of credible regulation in fostering a more reliable green bond market, which is essential for mobilizing the capital needed to address global environmental challenges.

This paper is structured in the following manner. First, I develop the theoretical framework, outlining the main hypotheses regarding how the new EU standard might influence green bond yields and the behavior of investors, particularly in the context of external verification and issuer environmental reputation. Then, I detail the empirical approach, including the dataset and the regression discontinuity in time (RDiT). The next section presents the results of the impact of the EU Green Bond Standard on green bond yields and explores the heterogeneous effects of external verification and issuer environmental reputation on green bond pricing, offering insights into how these factors interact with the new regulatory framework. Finally, I address how these negative short-term effects might be temporary and lead to a greening of the financial markets in the long run, before concluding and suggesting avenues for future research.

2 A Theory of the EU environmental leadership's impact on an asymmetric market

A central question in comparative political economy concerns the conditions under which public regulation meaningfully reshapes market behavior. This question is particularly salient in policy domains characterized by complex information, transnational markets, and extensive reliance on private governance. Environmental finance exemplifies these challenges. While markets increasingly channel capital toward ostensibly sustainable activities, the environmental quality of financial assets is often difficult to observe and verify, creating scope for strategic behavior and undermining market credibility

As the demand for sustainable and environmentally responsible investments grows, the green bond market has emerged as a pivotal tool for financing projects aimed at addressing climate change and promoting environmental sustainability. However, the rapid expansion of this market has also brought about challenges, particularly the risk of greenwashing. These concerns stem from the fact that the environmental quality of a green bond is not directly observable and must instead be inferred from disclosures, certifications, and reputational cues. In this section I develop a theory of how public environmental regulation can alter market outcomes by reshaping credibility and reputational incentives. I argue that when a legitimate public authority intervenes in a market characterized by heterogeneous private standards, regulation can have immediate economic effects even before full enforcement takes hold. Specifically, public regulation can reorder reputational stakes, increase short-term uncertainty, and induce more conservative market behavior as investors reassess the credibility of existing assets. Crucially, the argument does not hinge on regulation per se, but on the distinctive political authority of the European Union as a supranational regulator capable of combining legal enforceability, market size, and reputational legitimacy in ways that private governance can easily replicate.

I develop this argument in three steps. First, I explain why fragmented market-based governance struggles to resolve information asymmetries in environmental finance. Second, I show how the European Union's intervention as a credible supranational regulator alters investor beliefs by enhancing legitimacy, coordination, and enforcement expectations. Third, I theorize why these effects are heterogeneous, disproportionately affecting firms with weaker environmental reputations as investors rely on reputational heuristics to navigate regulatory transitions.

Traditionally, “greenwashing” consists of companies making false or exaggerated claims about the environmental soundness of their products by selectively disclosing information about their environmental credentials, without fully disclosing any negative information in order to construct a positive environmental image (Lyon and Maxwell 2011). In the context of green bonds, companies may label projects that are not environmentally sound or ‘green’ as compliant with green bond requirements by providing deceptive information about the use of proceeds of the bond during the issuance process.

Since the issuance of the first green bonds in 2007, several green bonds have been criticized for engaging in corporate greenwashing. For instance, in 2017, Repsol, a Spanish oil and gas company, issued a green bond to finance and refinance energy efficiency investment in their chemical and refinery facilities to reduce greenhouse gas emissions. While the goal of the bond was to reduce emissions, critics argued that it would lead to an indirect increase in emissions over time as it would extend its plants’ lifetimes (CBI 2017). In 2016, Mexico City issued a green bond to finance the construction of a new airport which faced criticism for its overall environmental impact, including the destruction of a unique ecosystem and high carbon emissions associated with its construction (Gonçalves 2021).

The potential for greenwashing is increased by the fact that, until recently, there was no single global definition of what a green bond is and no standardized supervision mechanism that ensures that bond proceeds are effectively funding green projects. Instead, governance of green bonds is largely decentralized and conducted by a variety of market institutions and investment standards, such as the Green Bond Principles, the Climate Bonds Initiative certification scheme, and third-party verification of individual green bonds. Notably absent from this governance regime are state institutions and government regulation. In this sense, a public regulatory framework could have a significant impact on the green bond market by establishing clear, enforceable and uniform standards and ensuring transparency and accountability in the green bond market.

Nonetheless, given that public governance is a late comer, we could also expect that a public regulatory framework could result in nothing more than 'hot air' as preceding private governance regimes may compete with public regulators to be adopted by market participants (Park 2018). In this regard, there is a robust strand of research that suggests that private governance initiatives and regulations can serve as a political strategy to preempt more stringent public regulations by crowding out the demand for governmental interventions (Kolcava, Rudolph, and Bernauer 2021; Lyon and Maxwell 2004; Malhotra, Monin, and Tomz 2019; Potoski and Prakash 2013; Rhein and Sträter 2021).

Theoretically, therefore, we could expect a public regulatory initiative to have a limited impact given that it would compete with a well-established private regulatory framework. Indeed, such arguments help explain why the European Commission came to the conclusion that the introduction of a European credit rating agency would add little value to investors given the information already existing (EPRS 2016). It also helps explain the little attention investors paid to the EU Taxonomy before the announcent.[1] Nonetheless, as I will argue, the fragmented and pluralistic nature of the private regulatory framework that exists creates a situation where information asymmetry is present and, as such, poses several kinds of legitimacy challenges. This fact, coupled with the need for investors to satisfy demand for true green assets (Guerrero-Villegas et al. 2018; Muñoz-Torres et al. 2019; Ortas, Gallego-Álvarez, and Álvarez 2019), opens the possibility for a public governance initiative to have a significant impact in the green bond market by effectively reducing the space for greenwashing.

2.1 Governing green bonds through markets: information asym-

1. For instance the first appearance of 2023 of the EU GBS in News Monitor App of Refinitiv Workspace, the London Stock Exchange Group's terminal to analyze real-time financial market data, occurred after the legislation was approved.

metry and legitimacy.

As I mentioned previously, green bond governance is fragmented as there is no single global definition of what a green bond is and no standardized supervision mechanism that ensures that bond proceeds are effectively funding green projects. As a result, the authenticity of a green bond is a function of invisible characteristics subject to asymmetric information, a situation where the parties involved in the market possess unequal levels of information (Bachelet, Becchetti, and Manfredonia 2019, p. 4). This, in turn, affects their credibility because the investors are unsure about the true 'greeness' of the firms issuing green bonds (Rodrik 1989).

Information asymmetry has always been a key issue in bond markets. For instance, it has been shown that the level of information asymmetry on credit risk is related to bond liquidity (Copeland and Galai 1983; Glosten and Milgrom 1985), and bond pricing (Khalil et al. 2019; Krueger, Sautner, and Starks 2020; Lu, Chen, and Liao 2010). But unlike credit risk where there is a prevailing credit rating practice, green bonds add an additional level of asymmetry by having to assure potential investors of their green credentials but such disclosure measures are difficult for investors to verify and often lacks standardization, leaving credibility concerns unresolved. (Hyun, Park, and Tian 2020). In this respect, Cartellier, Tankov, and Zerbib (2024) use a general equilibrium model to show that greenwashing can be mitigated by enhancing transparency to reduce information asymmetry.

Thus, reducing the information asymmetry of a bond's "greenness" is crucial for green bonds, especially considering that green bonds generally trade at a premium or "greenium", resulting in a reduced cost of borrowing relative to equivalent conventional bonds (Zerbib 2019). As such, it has been theorized that for green bonds to be effective they need to address the challenge posed by this case of information asymmetry because if bonds that are not truly green can claim the benefits of greenium without any added costs, green bonds would lose their credibility (Schmittman and Gao 2022).

To address these challenges, issuers of green bonds can disclose certain information – such as how proceeds are allocated or the level of environmental outcomes – to increase transparency for the purpose of signaling the bonds environmental commitments to potential investors, nonetheless credibility remains a concern because there are unclear motives about the issuing firms environmental credentials (Clarkson et al. 2019; Rodrik 1989). To circumvent this credibility barrier bond issuers have increasingly relied on verification by independent third parties to enhance their signaling communications with the relevant groups (Ehlers and Packer 2017; Simeth 2022). Of these independent reviews, second-party opinions (SPOs) issued by an independent research institution such as CICERO (recently bought by Standard and Poor's), ISS-Oekom, Vigeo Eiris, Moody's, and Sustainalytics, are the most popular (Dinh, Eugster, and Husmann 2023; Dorfleitner, Utz, and Zhang 2022). As such, the financial economics literature has increasingly focused on the impact of SPOs on green bond prices.

While there are some studies that find no significant relationship between external verification and green bond prices (Zirek and Unsal 2023), the majority of studies find that a credible and independent review is valuable to investors as a means of reducing information asymmetry regarding greenwashing as they are more willing to accept lower interest rates (i.e. returns) on green bonds with verification than those without relative to similar conventional bonds (Dorfleitner, Utz, and Zhang 2022; Flammer 2021; Hyun, Park, and Tian 2020; Simeth 2022).

However, while external verification can help reduce information asymmetry and mitigate concerns about greenwashing, its effectiveness is limited by the fact that different market institutions and participants compete for market adoption with varying definitions of what constitutes a green bond. For example, while many SPO providers only provide a binary qualification for bonds that rates them as being either green or not green, others provide a qualitative guide of the 'true greenness' of green bonds (CICERO, for instance, grades bonds into 'shades of green' indicating the potential environmental impact of the bond) (Dorfleitner, Utz, and Zhang 2022). Moreover, the lack of harmonized rules for reviewing green bonds and diverging definitions of green activities make it difficult for investors to effectively compare bonds with respect to their environmental objectives. Theoretically, then, the lack of unified standards may undermine the reliability of certification and can lead to green bond issuers engaging in green window dressing. Famously, both the Repsol and Mexico City Airport green bonds had been externally verified by SPOs.

Alternatively, empirical work looking at the characteristics of firms that ask for verification services has found that a firm that has a poor environmental reputation is more likely than a firm with a good reputation to ask for assurance services to enhance its reputation (Dinh, Eugster, and Husmann 2023; Hummel, Schlick, and Fifka 2019). Intuitively, if a firm has a strong environmental reputation, it would not seek external verification as it already considers itself a legitimate green bond issuer; conversely, firms that are poor environmental performers might require an external actor in order to look greener. This suggests that even verified bonds can be treated as suspect by investors

Given this segmentation and lack of consistency, critics suggest that this heterogeneity undermines the credibility and legitimacy of market-based governance; this fragmentation leads to incoherent or conflicting regulatory mandates and uncertainty among market participants. Moreover, since, as a general rule, private governance standards do not have the same enforcement mechanisms as public regulation, which can draw on the coercive authority of government, they also lack accountability (Park 2018). Furthermore, the broader literature on private sector certification has highlighted the limited capacity of such schemes to produce substantive change. These initiatives often function more as tools for "checklist compliance" than as meaningful regulatory mechanisms (Locke 2013), or they may be inadequate in low-governance contexts, where they are unable to 'fill the void' in the absence of robust institutional frameworks with standards of best practice (Bartley 2018). For investors relying on private certification as a guideline of environmental performance, this raises questions about the greenwashing potential and informational value of such labels.

This last point is important given that recent literature suggests that investors, especially those with ESG mandates, face scrutiny and reputational risk over the environmental legitimacy of their portfolios (Starks, Venkat, and Zhu 2017). Furthermore, survey evidence indicates that green bond purchases are influenced by the environmental credentials at issuance and post-issuance performance, reflecting investor preferences that are not purely pecuniary but reputational (Sangiorgi and Schopohl 2021). In this sense, credibility operates as a reputational filter: institutional investors aim to avoid holding assets that may later be revealed as greenwashed, as this could undermine their own reputation with their clients. This logic builds on earlier work on ethical investing and portfolio exclusion (Heinkel, Kraus, and Zechner 2001), and on Fama and French (2007) challenge to the classic view of investor rationality as solely return-driven.

2.2 Governing green bonds through state institutions: Climate leadership and the EU Taxonomy.

Given this background of market heterogeneity and lack of legitimacy that pose risks of greenwashing and reputational risk for investors, a public and unified standard has the potential to have a significant impact on the green bond market. As such, within the broader context of the European Green Deal – the EU's overarching strategy to become climate-neutral by 2050 – in February 2023 the European Parliament and the European Council agreed on a proposal for a European Green Bond Standard.

First, this initiative addresses the issue of legitimacy as the European Union has a long-established history of environmental leadership and, specifically, the European Green Deal enabled the EU to increase its sway on the international stage (Tobin, Torney, and Biedenkopf 2023). This leadership matters not only politically but also economically as it provides the

EU with legitimacy as a standard-setter capable of coordinating expectations in international markets. Secondly, the EU green bond standard addresses the segmentation in the green bond market by linking it with the EU taxonomy, a regulatory framework that clarifies what constitutes sustainable activities and what it means for transition, thus making it easier to identify what is a green project. The Taxonomy translates EU policy commitments for use by capital markets. As such, it bridges the gap between policy and investment practice by signalling which projects are consistent with its environmental objectives. Specifically, bonds certified under the EU green bond standard would need to ensure that at least 85% of the use of proceeds of the bond are allocated to economic activities aligned with the EU Taxonomy. Crucially, it also establishes the supervision of issuers by instituting mandatory ex-post verification of allocating reporting where firms are required to submit yearly reports demonstrating that the proceeds of the European Green Bond, from its issuance date until the end of the period referred to in the report, have been allocated in accordance with the regulation during the lifetime of the bond (TEG 2020).

In this sense, the EU Green Bond Standard was designed to overcome the biggest barriers to the green bond market outlined above. To address legitimacy issues, it builds on the EU Taxonomy to clarify what constitutes as 'green' to reduce controversies and reputational risk for issuers and investors. Secondly, to address the heterogeneity, it proposes a standardized verification process which is expected to streamline the process and reduce the cost of external reviews. It also lays the basis for policymakers to design policies and instruments to incentivize the issuance of green bonds (TEG 2020, pp. 11-12). As such, bond investors should have little doubt that if a green bond is certified under the EU, their use of proceeds are not being greenwashed. Thus, a systematic green bond standard issued by a legitimate environmental leader could reduce the heterogeneity of the green bond market and reduce the space for greenwashing. From a comparative perspective, this suggests that similar regulatory interventions are unlikely to generate comparable market responses in the absence of a political authority that combines credible enforcement with transnational legitimacy and agenda-setting power

Finally, given the discussion above on investors being wary of reputational risk over environmental legitimacy, the introduction of the EU Green Bond Standard (GBS) could, theoretically, increase the reputational risk associated with holding greenwashed assets. Under the status quo, the risk of public exposure for holding greenwashed green bonds was minimal as there was no widely accepted and robust standard for exposing greenwashed bonds. The GBS changes this by having a robust environmental standard. While in the long-term it would reduce the reputational risk of holding greenwashed assets, in the short term, however, the arrival of a credible public regulator is likely to induce more conservative investor behavior, leading to higher green bond yields immediately following the announcement of the legislation. As such, I theorize that a public green bond standard issued by a legitimate environmental leader such as the EU would lead to an increase in yields in the short term by reducing the heterogeneity of the green bond market.

2.3 Heterogeneous effects of public regulation in green bond markets.

As I argued in the last section, governing green bonds through market institutions leaves room for greenwashing as the issue of information asymmetry is not entirely resolved. As such, I theorized that a public green bond standard would have a significant impact in the bond market if it came from an actor seen as legitimate. However, to test whether this effect is successful in reducing the space for corporate greenwashing, we should expect that it is not homogeneous across the board. That is, the effect should be more pronounced for firms perceived to be engaging in greenwashing relative to those that are not.

As I mentioned previously, the financial economics literature focusing on green bonds addresses the question of greenwashing mainly by analyzing the effects that third-party verification of individual securities has on their pricing dynamics as effective signaling tools of their environmental credentials (Dinh, Eugster, and Husmann 2023; Dorfleitner, Utz, and Zhang 2022; Ehlers and Packer 2017; Simeth 2022). This body of work highlights how internal characteristics of bonds can shape their pricing dynamics. Yet this literature does not tell us much about how the differences across issuers – beyond economic differences – matter. For instance, why an oil and gas company (i.e. Repsol) can exploit the benefits of a green bond premium by having a green bond with external verification.

Political economy research shows that environmental regulation tends to impose higher costs on firms that are more carbon-intensive or environmentally controversial, while firms with lower adjustment costs may benefit from stricter standards. For example, energy-intensive industries in Germany resisted emissions trading due to high compliance costs Meckling (2015). Some firms may support regulations if they have lower adjustment costs, aiming to disadvantage competitors with higher costs and increase market share Kennard (2020). Similarly, workers in polluting industries are less supportive of climate cooperation, fearing higher economic costs Bechtel, Genovese, and Scheve (2019). Sector characteristics shape firm responses to regulation, with high emitters and vested interests often resisting or delaying climate policies, while industry leaders can drive constructive policy feedback Genovese and Tvinnereim (2019), Kim, Urpelainen, and Yang (2016), and Brulle (2021).Overall, the literature converges on the notion that the this leader–laggard distinction suggests that firms' pre-existing environmental positions shape their exposure to regulatory change.

Beyond objective compliance costs, investor decision-making under uncertainty further amplifies these differences. While green bond standards are intended to reduce information asymmetries by providing credible and comparable assessments of environmental performance, investor decision-making often remains shaped by broader factors that go beyond the technical content of individual securities. In this sense, there is a robust body of works that shows that in contexts of uncertainty and limited verifiability, investors frequently rely on heuristic devices, such as country classifications, credit ratings or affiliations with reputable international actors, to simplify complex evaluations (Brazys and Hardiman 2015; Brooks, Cunha, and Mosley 2015; Gray 2013; Mosley 2003). These heuristics are not merely cognitive shortcuts but they actively shape capital flows, interest rates, and perceptions of risk, sometimes leading to contagion or inefficiencies disconnected from economic fundamentals (Brooks, Cunha, and Mosley 2015). The implication here is that reputational context matters and that firms with strong environmental reputations may face fewer questions about the true 'greenness' of their bonds and, in contrast, "laggard" firms may face greater scrutiny. [2]

In this way, we can think that the leader-laggard distinction can operate as a reputational heuristic in investors' assessments of green bonds and as a potential mechanism behind their decision on whether to hold a bond to maturity or not. While investors may not have the capacity or the inclination to read the fine print of every bond they are buying (Choi et al. 2024), they mayrely on heuristic devices—such as firm reputation, ESG ratings, past controversies and similar metrics provided by financial market data and analytics platforms such as Bloomberg and the London Stock Exchange Group to assess risk. These heuristics function as reputational shortcuts, allowing investors to infer the likelihood that a bond may later be exposed as greenwashed without scrutinizing technical details.

In this context, issuer reputation becomes a key mechanism linking public regulation to heterogeneous market outcomes. Firms with strong environmental reputations are more likely to be perceived as credible green bond issuers and therefore face less scrutiny when standards tighten. By contrast, firms with poor environmental records are more likely to be viewed as potential greenwashers. When a credible public standard such as the EU GBS

2. Here I emphasize that this classification of issuers as "leaders" or "laggards" applies only to the subset of firms that issue green bonds. It does not extend to all firms in the broader financial market. I recognize that firms opting into this market already reflect a degree of self-selection, however, this approach is consistent with the paper's focus on secondary market dynamics among participating issuers rather than on primary market dynamics that determine whether a firm chooses to issue a green bond or not.

is introduced, investors may disproportionately reassess bonds issued by these laggards, demanding higher yields to compensate for perceived reputational and regulatory risk.

Thus, the leader–laggard distinction operates not only through material compliance costs but also through reputational heuristics that shape investor responses to regulatory change. This logic implies that in the short-term period after the announcement of the EU GBS there will be an adjustment period that has stronger pricing effects for bonds issued by environmentally weaker firms. During this transition period, investors face heightened uncertainty about which existing bonds and issuers will ultimately be perceived as compliant with the new standard. In such contexts, investors are more likely to fall-back on reputational heuristics—such as issuer-level environmental scores and past controversies—as provisional signals of credibility. Therefore, I theorize that the EU green bond standard will have a more pronounced impact on firms with poor environmental reputations.

2.4 Hypotheses

In sum, I argue that the EU Green Bond standard, despite entering a well-established market late, will significantly influence green bond pricing. Given market heterogeneity and information asymmetry, a unified standard from a credible source would lead to higher yields for green bonds, as investors would become more cautious and penalize environmental controversies, addressing corporate greenwashing.

However, this effect will not be uniform. Firms classified as climate laggards, or those with lower environmental scores, will likely face higher yields due to the new standard. Based on this, I propose the following main hypotheses.

- **H1:** It will be more expensive to borrow through green bonds after the announcement of EU legislation.

- **H2:** Green bonds issued by firms with poor environmental scores (i.e. laggards) will experience higher yields than those issued by firms with higher environmental scores (i.e. leaders).

3 Data and empirical approach

To assess the empirical implications of the theoretical discussion, I focus on daily green bond yields as the primary outcome variable. Yields reflect the interest investors earn from holding bonds until maturity, with higher yields indicating greater returns for investors but higher borrowing costs for issuers. Since bond yields and prices are inversely related, a rise in yields corresponds to a drop in bond prices. I collected data on all active bonds traded between February 2022 and May 2023, along with daily yield observations, from Refinitiv, the London Stock Exchange Group's (LSEG) data provider.

Moreover, from Refinitiv I also collected several measures on issuers' environmental credentials to test for heterogeneous effects of the EU Green Bond Standard as a proxy for environmental leaders and laggards. First, I collect ESG scores, an overall score based on self-reported information relative to environmental performance and capacity. ESG scores measure the issuer's ESG performance based on public data. I also extract the ESG Controversies score which measures a company's exposure to environmental controversies and negative events reflected in the media. As such, during any given year, if a scandal occurs, the company involved is penalized and this affects their overall Controversies score. The impact of the event may still be seen in the following year if there are new developments related to the negative event, for example, lawsuits, ongoing legislation disputes or fines. It is important to note that both measures account for industry and company size/market capitalization biases as larger companies and issuers from certain industries attract more media attention.

Refinitiv grades issuers based on a letter scale that ranges from A+ to D- on issues such as the firm's ESG score and its ESG controversies score where those with grades on the higher end of the spectrum are considered ESG leaders and those with grades on the lower end of the spectrum are considered ESG laggards. These grades are useful given that, they can serve as useful heuristic devices to which investors are likely to respond given that they provide simple summary assessments of firms' ESG ratings, are comparable across cases, and can be easily incorporated into their financial analyses (Mosley 2003).

To estimate the treatment effect of EU regulatory policies on green bond yields, I rely on a regression discontinuity in time (RDiT). A typical regression discontinuity design consists of a running variable or score, a cutoff score, and a discontinuous treatment assignment rule where all units are assigned a value of the running variable and treatment is assigned to those units whose value exceeds the known cutoff score and all others remain untreated such that for a treatment variable X_i observed for each i, if $X_i > c$, the unit is treated, and if $X_i < c$, the subject is not. The main intuition behind this is that units observed near either side of the threshold are very similar so we can assume that nothing else affects the discontinuity in the outcome apart from the treatment. The mild assumptions associated with these designs have made them expand rapidly in the context of policy evaluation and have made these designs one of the most credible non-experimental methods for policy evaluation and causal inference. In contrast to a typical RD design, an RDiT design the cutoff c consists of the date of a policy change instead of a score so that for all dates $t > c$, the units are considered treated and for all other dates $t < c$, the units are not (Cattaneo and Titiunik 2022; Hausman and Rapson 2018).

The local average treatment effect (LATE) is estimated using a non-parametric approach that fits observations on a low-order polynomial separately for treated and control groups. This estimation focuses only on observations near the cutoff, rather than using all available data. A triangular kernel function is applied to give more weight to observations closer to the cutoff, and a bandwidth is chosen to define the effective sample size, minimizing the asymptotic mean squared error (MSE).

This non-parametric approach has become the standard choice for RD designs as using parametric approaches we would have to get the functional form of the polynomial right to address any bias introduced by non-linearities between the running variable and the outcome variable. Instead, by using this approach we do not make assumptions about the underlying data or the functional form of the polynomial; rather, the data drives the shape and produces an estimator with robust bias-corrected standard errors, and p-values (Calonico, Cattaneo, and Titiunik 2014).

In this context I leverage the complexity of the EU's legislative process which has been described as Byzantine and ineffective in that it systematically generates extremely watered-down deals that hardly change existing European or national provisions (Crombez and Hix 2015), to use as an as-if random treatment. A potential concern in such designs is the presence of anticipation effects, whereby political events are priced in before they occur. However, as Bechtel and Schneider (2014) argue, such concerns are mitigated in the context of EU policymaking. Specifically, they note that the precise agenda of European Council meetings is frequently unknown beforehand, making it difficult for investors to predict the timing and content of policy announcements. Furthermore, the EU's "stop-and-go" pattern of integration complicates expectations, even if the agenda were known. This combination of institutional opacity and policy unpredictability supports the assumption that market participants cannot systematically anticipate outcomes, thus reinforcing the plausibility of my identification strategy.

The European Council and European Parliament reached a political agreement on the EU Green bond Standard on the 28th of February 2023; however, this agreement was announced the following day on the 1st of March. As such, I will use that date as the cutoff date (c).

The outcome variable is the daily yield of individual green bonds. The empirical strategy estimates whether bond yields exhibit a discontinuous change at the announcement date. Intuitively, if investors reassessed regulatory and reputational risks following the announcement, this should be reflected in an immediate shift in yields at the cutoff.

To test the main hypothesis, I follow the the nonparametric approach of CCT (2014) and estimate the LATE using a local linear regression fitted separately on either side of the threshold to avoid the inference problems that characterize higher-order polynomials such as overfitting the data, producing noisy estimates, and poor coverage of confidence intervals, as well as the fact that higher-order polynomials are harder to justify and interpret theoretically (Cattaneo and Titiunik 2022; Gelman and Imbens 2019).

The running variable is time, measured in trading days relative to 1 March 2023. Estimation is restricted to observations within a data-driven bandwidth around the cutoff and applies a triangular kernel that assigns greater weight to observations closer to the announcement.

In the main specification, I employ a local linear polynomial $p = \{1\}$. However, following Zhuan Pei and Weber (2022) who observe that the best polynomial ranges from local constant to quartic, I provide alternative estimations with polynomials $p = \{2, 3, 4\}$ as robustness checks. Following Calonico, Cattaneo, and Titiunik (2014), I specify a mean-squared optimal bandwidth of 33 days around the cutoff date (Hausman and Rapson 2018) and also provide alternative specifications with different bandwidths to test the robustness of the main estimate. In addition to this, I include a vector of time-varying bond characteristics that serve as controls, such as liquidity and time-to-maturity, as well as issuer and daily fixed effects to account for any unobserved heterogeneity, and standard errors are clustered at the individual bond level.

Furthermore, to test for the heterogeneous effects specified by H_2which refer to verification and environmental reputation status respectively, I follow an analogous model to (1) where I pool all the bonds across the Refinitiv Grades and condition them on the grade the issuing firms received based on the definitions discussed above.

4 Results and discussion

4.1 Results.

In this section I present the results of the empirical approach outlined in the previous section. First, in Figure 1, we see a regression discontinuity plot (RD Plot) of the full data set that illustrates the relationship between bond yields and days relative to the approval of EU Green Bond Standard. The individual data points plotted in gray, representing the daily means of green bond yields. The blue line, which represents a polynomial function fit to the data, is fitted separately on either side of the vertical dashed line marking the cut-point. As can be clearly seen in the plot, green bond yields show a substantive increase after the announcement of the Green Bond Standard with a visible discontinuity in the cutoff point. However, the visual evidence alone does not confirm whether this discontinuity is statistically significant, nor does it account for any underlying factors that might differ across the cut-point. While RD plots are useful to provide a visual guide of the research design, it should not be used as a substitute for the formal estimation of discontinuity effects (Cattaneo and Titiunik 2022).

The formal RDiT estimates are done following the nonparametric approach of CCT mentioned previously and appear in Figure 2. This figure shows the point estimates and 95% confidence intervals of the local average treatment effect that euro-denominated green bonds exhibited after the EU Green Bond Standard was announced. Following CCT I present three estimates: a traditional linear estimation without adjusting for potential bias in the polynomial specification as mentioned above, a bias-corrected point estimate that accounts for the asymptotic bias introduced by the curvature of the regression function, and an additional bias-corrected point with robust standard errors.

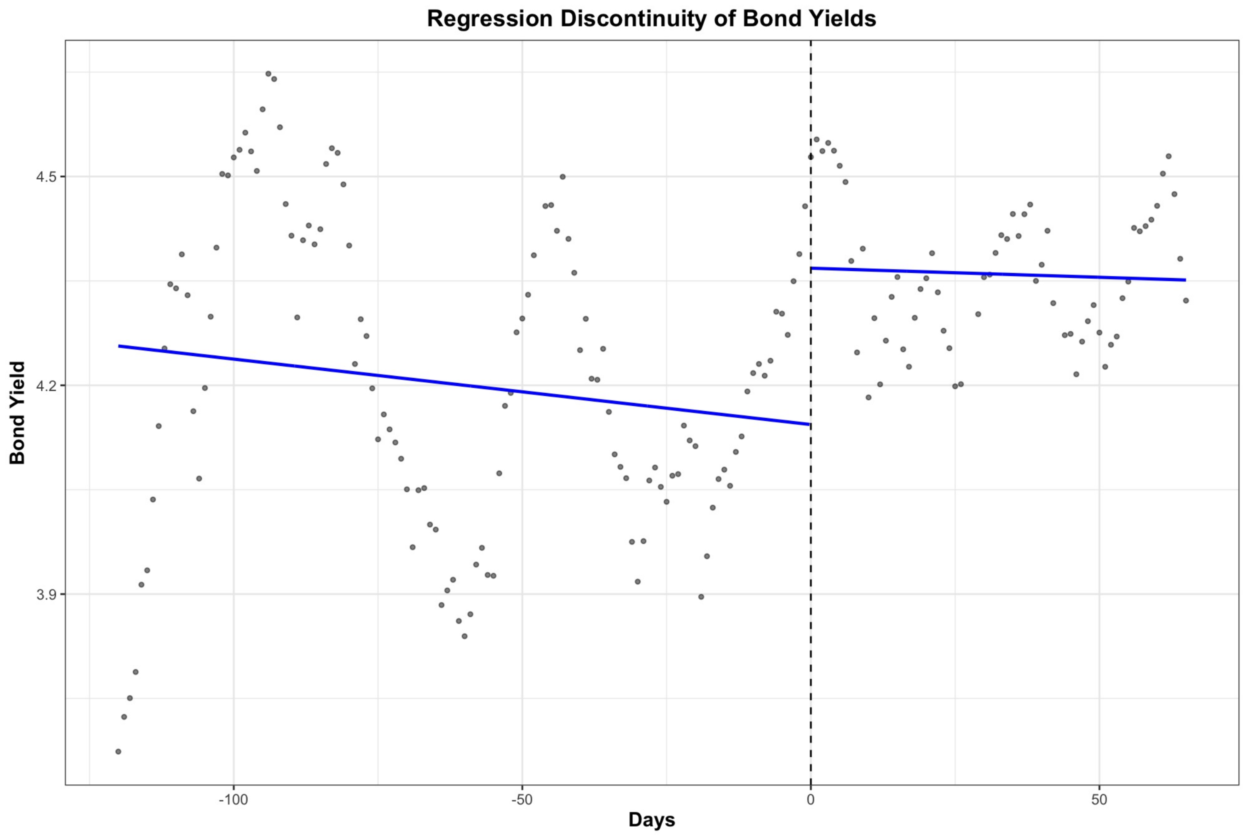

Figure 1: RD Plot of Green Bond Yields by issuer

Figure 2 illustrates a substantive and statistically significant increase in green bond yields following the EU legislation. The figure also shows that a traditional estimate, without bias-correction, is more conservative, indicating only a 15 basis points (bps) increase. In contrast, both the bias-corrected and robust estimates suggest an increase closer to 19 bps. To put this effect in context, Central Banks typically adjust interest rates in 25 bps increments when attempting to slow down the economy so we can think of the announcement of EU Green Bond Standard as having a comparable slowing-down effect in green bonds. In the previous section I theorized that this increase is due to the fact that bond investors are wary of greenwashing and, as such, a new credible standard that reduces information asymmetry in the market would make the bond investors act more conservatively. This slow-down of the green bond market after the EU announcement shows robust initial support for H_1. The RDiT estimates for the plot appear in Table 1.

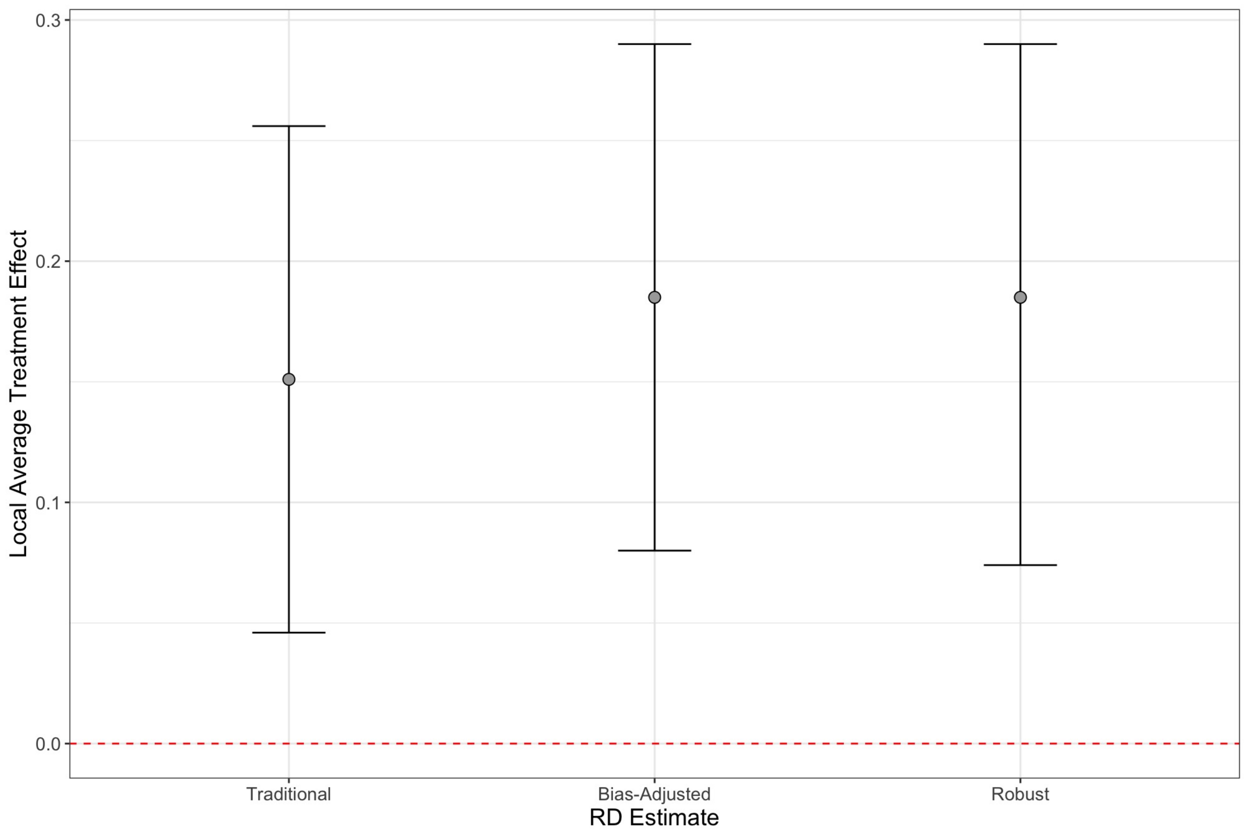

Figure 2: Local Average Treatment effect

Secondly, I turn to how differences across issuers matter by looking at issuers that can be thought of as environmental leaders and those that can be classified as environmental laggards. Figure 3 plots the two different proxies discussed above for environmental leaders and laggards, namely environmental controversies and ESG scores. For brevity, I present only plots for the robust estimators of the two analyses, but the full plots are available in the Supplementary Material. The first panel shows the coefficients and confidence intervals of the firms divided by their ESG Score. Conversely, the second panel shows the grades of issuing firms based on their score on environmental controversies. As can be seen in both cases, bonds issued by firms with grades A and B saw increase of yields of between 7 and 14 bps and these coefficients are not statistically significant. On the other hand, those with grades of C or D saw significantly higher yields (between 19 and 33 bps) both substantively and statistically than those at the 'leader' side of the spectrum.

Outcome	Sample	Method	RD Estimate	SE	P-Value
	Full	Conventional	0.151	0.054	0.005
Bond Yields		Bias-Corrected	0.185	0.054	0.001
		Robust	0.185	0.057	0.001

Table 1: RD Estimates by Type of Issuer

Note: RDiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel, and Mean Squared Error optimal bandwidth. Using issuer, industry, country, and daily fixed effects and controlling for liquidity (Bid-Ask spread) with cluster-robust standard errors at the ISIN level.

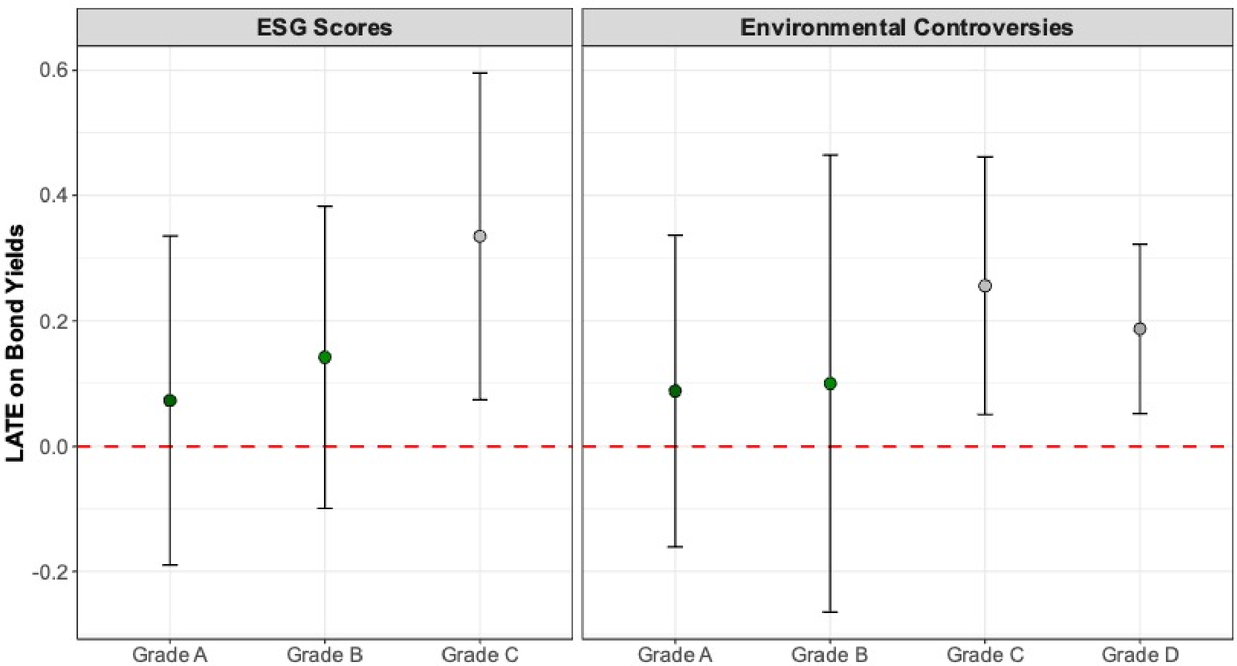

Figure 3: LATE on Bond Yields for leaders and laggards

Figure 3 and the respective estimates in Table 2 show marked heterogeneity and provide suggestive evidence that environmental reputation serves as a potential mechanism behind the local treatment effect observed after the announcement of the green bond standard in bond yields. Moreover, the point estimates are larger for bonds issued by environmental laggards, but their confidence intervals are smaller and are the only ones that are statistically significant. Together, I take these estimates to provide strong support for H_2 given that the increase of yields was more pronounced on issuers that are seen as environmental laggards.[3]

Table 2: RD Estimates by ESG Sample, Grade, and Method

Outcome	Sample	Subsample	Estimate Type	Coef	SE	p-value
Bond Yields	ESG Controversies	Grade A	Traditional	0.189	0.121	0.118
		Grade A	Bias-Adjusted	0.088	0.121	0.465
		Grade A	Robust	0.088	0.127	0.488
		Grade B	Traditional	0.216	0.182	0.236
		Grade B	Bias-Adjusted	0.100	0.182	0.583
		Grade B	Robust	0.100	0.186	0.591
		Grade C	Traditional	0.261	0.096	0.007
		Grade C	Bias-Adjusted	0.256	0.096	0.008
		Grade C	Robust	0.256	0.105	0.014
		Grade D	Traditional	0.123	0.066	0.064
		Grade D	Bias-Adjusted	0.187	0.066	0.005
		Grade D	Robust	0.187	0.069	0.007
Bond Yields	ESG Scores	Grade A	Traditional	0.163	0.128	0.202
		Grade A	Bias-Adjusted	0.073	0.128	0.566
		Grade A	Robust	0.073	0.134	0.583
		Grade B	Traditional	0.209	0.112	0.061
		Grade B	Bias-Adjusted	0.142	0.112	0.205
		Grade B	Robust	0.142	0.123	0.249
		Grade C	Traditional	0.307	0.120	0.010
		Grade C	Bias-Adjusted	0.335	0.120	0.005
		Grade C	Robust	0.335	0.133	0.012

Note: RDiT estimates of the local average treatment effect (LATE) on bond yields by ESG grade. Estimates use first-order polynomials, triangular kernel, and MSE-optimal bandwidth. All regressions control for issuer, industry, country, and day fixed effects, as well as liquidity (bid-ask spread). Standard errors are clustered at the ISIN level.

In the previous section, I suggested that if it were the case that the credibility of the EU Green Bond Standard would have an impact due on the asymmetry in the green bond market, the effect shown in the previous estimate would not be homogeneous but would differ depending on the environmental credibility of the bonds.

In contrast to an issuer's environmental reputation, Figure 4 shows the regression discontinuity estimates of the subgroup of bonds with a second-party opinion and those without one and the formal estimates are shown in Table 3. While there are some slight differences,

3. No firm included in the sample received a D grade in their ESG score.

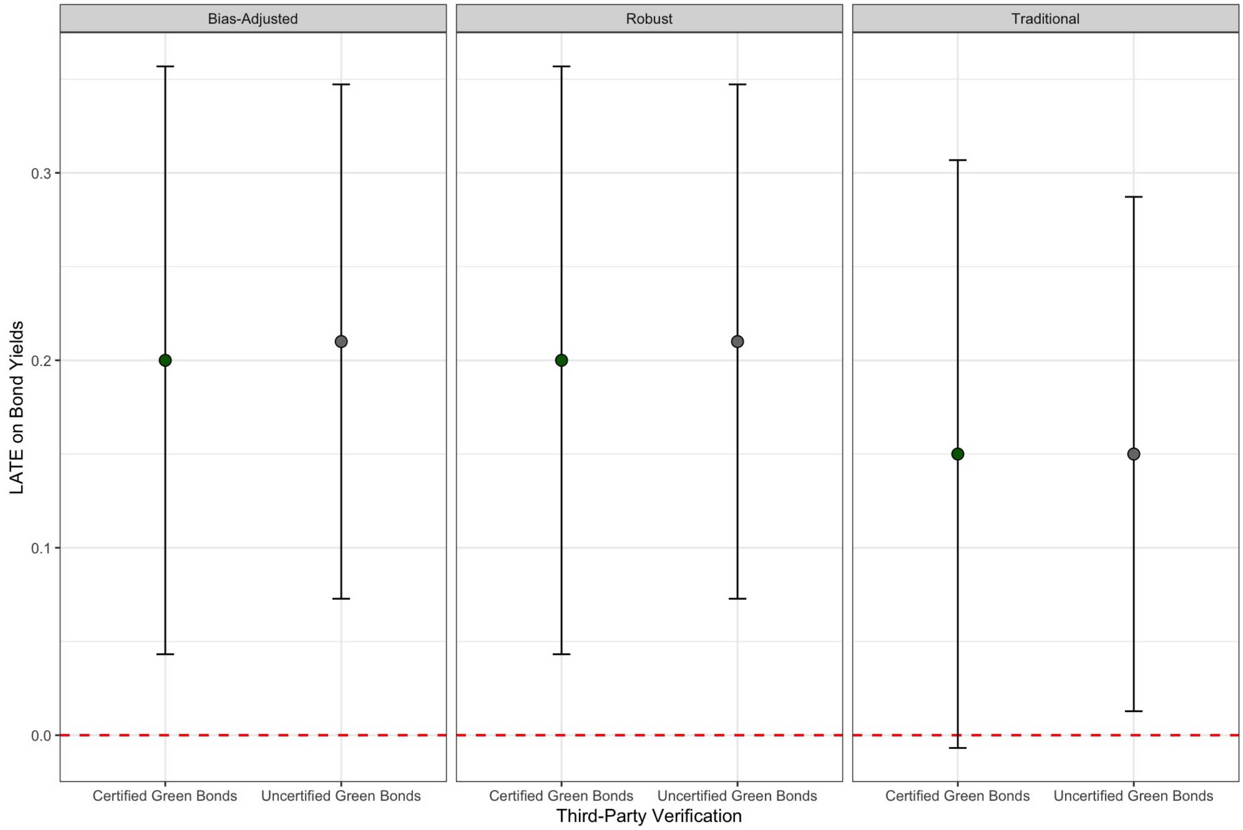

Figure 4: LATE SPO vs Non-SPO

these appear to be neither substantively nor statistically significant. These results and the theory outlined above suggest that the heterogeneity in the verification market is not enough to effectively reduce the information asymmetry of green bonds. Again, controversial cases such as airports or fossil fuel companies receiving external verification support the theory.

Robustness Checks To increase our confidence in these results I conducted a series of additional robustness tests and found that they are highly robust to other methodological approaches. A full discussion of these tests, appears in the Supplementary Material, however, it should be noted that these robustness checks indicate that the cutoff point is meaningful through a series of regressions with placebo cutoff points 5 trading days before and after the specified cut-point (Figure B.1) and the results presented here do not depend significantly on the specified polynomial specification as the results are robust for polynomials

Table 3: RD Estimates by Third Party Certification

Outcome	Sample	Method	RD Estimate	SE	P-Value
Bond Yields	Certified	Conventional	0.150	0.080	0.050
		Bias-Corrected	0.200	0.080	0.010
		Robust	0.200	0.080	0.010
	Not Certified	Conventional	0.150	0.070	0.030
		Bias-Corrected	0.210	0.070	0.000
		Robust	0.210	0.070	0.000

Note: RDiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel, and Mean Squared Error optimal bandwidth. Using issuer, industry, country, and daily fixed effects and controlling for liquidity (Bid-Ask spread) with cluster-robust standard errors at the ISIN level.

$p = \{1, 2, 3, 4\}$(Figure B.3) and different bandwidths (Figure B.4). Most importantly, I test the design using a series of placebo outcomes such as Euro-denominated conventional bonds and Dollar-denominated green bonds and find that the announcement of the EU Green Bond standard had no discernible effect on their yields (Figure A.1). To support my hypothesis that public standards have a much larger effect on bond pricing that private ones, I analyzed the main effect using an alternative analysis where I estimate the effect of the Climate Bond Initiative Standard (a private regulatory standard) on Green Bonds yields and find no effect (Table A.5). This lack of effect of the EU legislation on bonds other than Euro Green bonds, when combined with the results from the main RDiT estimates, and the lack of noticeable effect of analogous private standards suggests a causal effect where public regulation leads to a more conservative approach from investors towards green bonds. Finally, I assess the effect of the Green Bond Standard (GBS) on the primary market to determine whether it influenced bond issuance volumes. I find no meaningful difference in the amount of issuance before and after the announcement (Table A.3 and Table A.4). Crucially, the observed effect appears to be concentrated among issuers with poor environmental reputations. Overall, these results and the theory outlined above imply that EU regulation was effective in curbing greenwashing in the green bond market in the short term, largely due to the heterogeneity and information asymmetry inherent in market-based definitions of what qualifies as 'green'.

4.2 Discussion.

The results presented here paint a clear picture of the immediate effects that the introduction of the EU Green Bond Standard (GBS) has had on the green bond market. In the short term, as demonstrated by the regression discontinuity analyses, the regulation prompted a noticeable shift in investor behavior. This shift occurred through a conservative repricing of green bonds, especially among issuers with weaker environmental reputations. While expected, this reaction highlights the importance of credibility in environmental claims within financial markets and offers insight into how state regulation can reshape market dynamics. More broadly, the findings highlight how supranational regulation can reshape market behavior not only through formal compliance mechanisms, but by reconfiguring reputational incentives in policy domains characterized by information asymmetry and extensive reliance on private governance.

These short-term effects suggest that investors respond quickly to formal regulations and place greater emphasis on issuer credibility once new standards are introduced. The market's immediate reaction implies that future regulatory initiatives could benefit from increasing transparency and standardization to reduce information asymmetry and mitigate greenwashing. As my theory suggests, the pre-existing reliance on market-based mechanisms and private standards created fragmented definitions of what qualified as green. This heterogeneity allowed firms to operate under minimal scrutiny. The mixed evidence on the role of external verification, illustrated in Figure 3, further shows the limitations of private certification. In this context, the EU's credibility as a climate policy leader played a central role. The announcement of the GBS sent a strong signal to the market, and the observed reaction likely reflects investors' belief in the added value of EU regulation. If the regulation had lacked credibility, or if the EU had proposed a financial instrument outside its perceived domain of competence, such as a credit rating standard, a market response of this magnitude would have been unlikely.

Beyond these immediate effects, the findings raise important questions about the longer-term consequences of public regulation in sustainable finance. A key issue is whether this initial market response will evolve into sustained differences in bond valuation, or even incentivize improvements in environmental performance among issuers. In the short term, the GBS results in higher yields as a across the board as the market adjusts to new, more stringent requirements. As investors become more selective and the market matures under the new framework, a clearer distinction between credible green issuers and laggards is likely to emerge. Over time, this could lead to a two-tiered market in which firms that meet the GBS and have strong environmental scores enjoy a higher greenium, while issuers with poor environmental scores face reduced access to green capital or higher borrowing costs. This is especially true in the EU given its track history of influencing climate outcomes through convergence and policy diffusion (Gawel and Strunz 2019; Bayer and Aklin 2020)

Although the GBS may increase costs for some issuers in the short term, it could bring substantial benefits to the green bond market over the long run. These benefits may occur through mechanisms associated with the California effect and the Porter Hypothesis. Both frameworks suggest that strong regulation can drive meaningful behavioral change by creating financial or competitive incentives for firms to comply. The California effect, first conceptualized by David Vogel (Vogel 1995), explains how firms adopt stricter standards to access valuable markets. This phenomenon has been documented across environmental and labor domains (Distelhorst and Locke 2018; Konisky 2007; Malesky and Mosley 2018; Prakash and Potoski 2006). In the case of green bonds, the GBS could function similarly by pushing issuers in less-regulated contexts to align with EU standards in order to benefit from investor demand for certified green assets. As the green bond market grows, this may lead environmental laggards to improve their practices to regain competitiveness and gain access to the greenium.

As this process unfolds, the short-term reduction in the greenium reflects investor caution given an increased risk of public exposure for holding greenwashed assets. However, this period of adjustment may eventually result in the bifurcation of the green bond market. Issuers with higher environmental scores would be rewarded with lower yields while bonds from laggards would face higher yields. In the long term, this distinction could drive broader improvements in environmental performance by reinforcing the financial advantages of credibility and compliance.

The Porter Hypothesis offers a complementary explanation of how environmental regulation can lead to innovation and productivity gains in the long term while having a negative impact in the short term. While conventional economic theory suggests that regulation imposes costs and reduces efficiency, Porter argues that well-designed policies can actually increase competitiveness. In the context of green bonds, the GBS may encourage firms to invest in new technologies or sustainable practices to meet the standard, thereby enhancing long-term financial and environmental performance and, over time, these investments can yield returns through an increase in the greenium of the bonds they issue. This aligns with evidence from previous studies that find the immediate negative effects of regulations on productivity are temporary and that in the long run, regulation leads to higher levels of productivity compared to firms in less regulated areas (Berman and Bui 2001; Lanoie, Patry, and Lajeunesse 2008).

Thus, while the short-term effect of the GBS is a decline in greenium due to heightened scrutiny and investor uncertainty, the regulation has the potential to generate a more credible and efficient green bond market over time. This shift would reward transparency and environmental responsibility while penalizing greenwashing and would be in line with conventional financial theory that suggests that the introduction of the GBS could reduce risk and therefore lower yields. If the long-term dynamics unfold as the theory suggests, the GBS may not only improve market functioning but also contribute to environmental progress by aligning financial incentives with sustainability goals.

5 Conclusion

The analysis presented in this paper demonstrates that the introduction of the EU Green Bond Standard (EU GBS) has had a discernible impact on the green bond market, particularly in the pricing dynamics of these financial instruments. The empirical evidence suggests that the announcement of the EU GBS led to a significant increase in green bond yields, reflecting a shift in investor behavior towards greater caution and scrutiny. This finding supports the hypothesis that the introduction of a credible and stringent regulatory standard can influence market perceptions and drive a more conservative pricing approach, particularly in markets characterized by high levels of information asymmetry, as seen in the green bond market.

One of the key insights from this study is the role of environmental reputation in mediating the impact of the EU GBS. Bonds issued by companies classified as environmental laggards experienced a more pronounced increase in yields compared to those issued by environmental leaders. This suggests that the EU GBS has the potential to exert pressure on issuers with weaker environmental credentials, incentivizing them to improve their practices to remain competitive in the green bond market.

In conclusion, the introduction of the EU Green Bond Standard represents a significant step forward in the development of the green bond market. While the short term market response has been characterized by increased yields, particularly for issuers with weaker environmental reputations, this should be seen as a positive development in the long term as the EU GBS can help foster a more robust and credible market for green bonds, ultimately contributing to the global effort to finance the transition to a sustainable economy. As the market continues to evolve, ongoing research and monitoring will be essential to assess the long-term impacts of the EU GBS and to ensure that the green bond market remains a powerful tool in the fight against climate change.

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Supplementary Material for: EU Regulatory Power, Market Credibility, and the Political Economy of Green Finance

Contents

Appendix A. Placebo Outcomes **3**

A.1: LATE on Placebo Outcomes 3

A.2: RD Estimates using Placebo Outcomes 4

Effects on Primary Market 5

A.3: LATE on Issue Size 5

A.4: Difference-in-Means 6

A.5: Effects of Private Regulatory Standards 7

Appendix B. Robustness Tests **8**

Placebo Cut Points 8

B.1: LATE Using different cutdates 8

Alternative Polynomial Specifications 9

B.2: LATE Using different polynomials 9

B.3: RD Estimates with different polynomials 10

Alternative Bandwidths 11

B.4: LATE Using different polynomials 11

Full Figures of Leaders and Laggards Estimates 12

B.5: Leaders and Laggards:ESG Scores 12

B.6: Leaders and Laggards: Environmental Controversies 13

B.7: Leaders and Laggards: Taxonomy Alignment 14

Appendix C. Alternative Specifications **15**

C.1:Environmental Leaders and Laggards 15

C.2: RD Estimates of Leaders and Laggards 17

Appendix A. Placebo Outcomes

In order to increase our confidence that the observed changes in bond yields are truly attributable to the Green Bond Standard, rather than other extraneous factors, I test the effect of the treatment on several placebo outcomes. This strategy helps control for confounding factors and also aids in ruling out placebo effects, thereby distinguishing between genuine treatment effects and random effects. Furthermore, assessing placebo outcomes enhances the internal validity of the study by providing evidence against alternative explanations for the observed results. It can also reveal the specificity of the intervention's effects, clarifying whether it has a targeted impact on the intended outcome or a broader, non-specific influence.

As such, I test the effect of the approval of the EU Green Bond Standard on the yields of Euro-Denominated Conventional Bonds and Dollar-Denominated Green Bonds, with the expectation that these samples should not experience any significant effects. Additionally, I also test the effect on liquidity, given its role as our main time-variant covariate.

As can be seen in both Figure A.1 and Table A.1, none of the placebo outcomes show substantive or statistically significant effects after the EU's announcement.

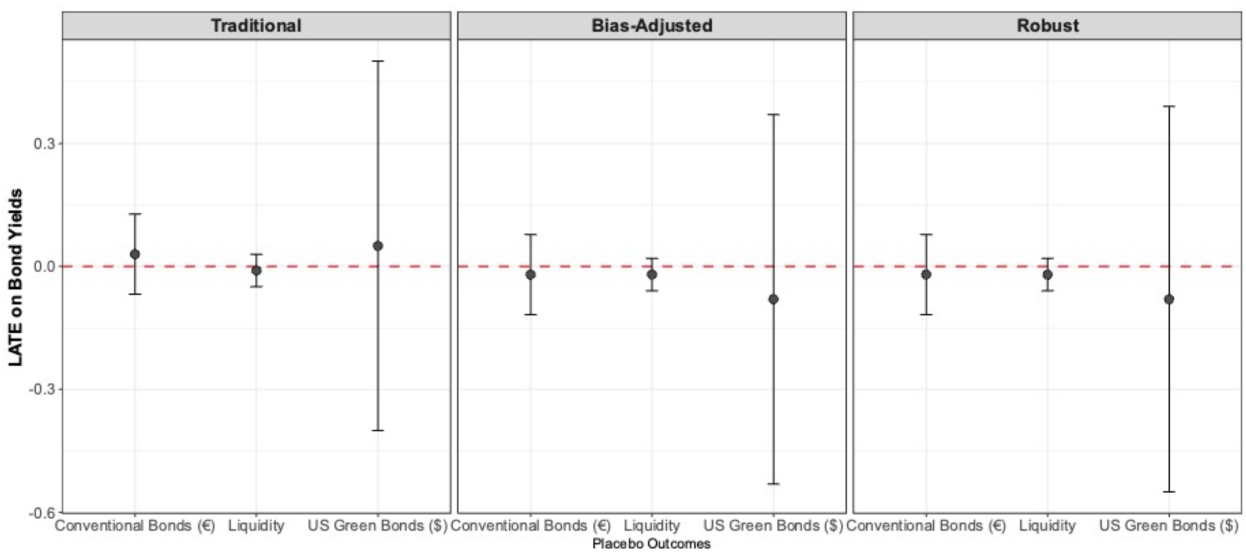

Figure 1: LATE on Placebo Outcomes

Table 1: RD Estimates using Placebo Outcomes

Outcome		RD Estimate	SE	P-Value
Green Bonds ($)	Conventional	0.05	0.23	0.83
	Bias-Corrected	-0.08	0.23	0.73
	Robust	-0.08	0.24	0.73
Conventional Bonds (€)	Conventional	0.03	0.05	0.50
	Bias-Corrected	-0.02	0.05	0.62
	Robust	-0.02	0.05	0.61
Liquidity (€ Green Bonds)	Conventional	-0.01	0.02	0.79
	Bias-Corrected	-0.02	0.02	0.38
	Robust	-0.02	0.02	0.41

Note: RdiT estimates, standard errors, and p-values of the Local Average Treatment Effect on bond yields for dollar-denominated green bonds issued in the US and for conventional bonds issued in the EU. Estimates with 1st order polynomials, triangular kernel, and Mean Squared Error optimal bandwidth using cluster-robust standard errors at the ISIN level.

Effect on Primary Market Supply

Given my focus on secondary market analysis, supply-side considerations that could influence primary market issuance in the immediate lead-up to the launch of the EU Green Bond Standard (EU GBS)—such as the possibility that environmentally weaker firms might withdraw from the market if they anticipate any impact from the EU GBS—are likely to have limited real-world implications. Furthermore, since I theoretically argue that yields would increase immediately following the introduction of the EU GBS, if environmentally lagging firms were to refrain from issuing green bonds, we would, in theory, expect to observe the opposite effect of what is reported.

Nonetheless, to address the potential concern that a discontinuity in yields might result from weak issuers choosing not to issue bonds due to expectations surrounding the EU GBS, I examine the effect of the Standard on primary market issuance. Specifically, I test whether there is a significant difference in the average size of bonds issued on either side of the EU GBS threshold. I first apply a Regression Discontinuity in Time (RDiT) approach, similar to that used in the main analysis using the issue size of green bonds (logged) issued either side of the cutoff date

I then compare the average bond issuance between the two-month periods before (January–February 2023) and after (March–April 2024) the announcement of the Standard using a two-sample t-test.

As shown in Tables A.2 and A.3, the EU GBS has no discernible effect on primary market outcomes.

Table 2: RD Estimates of GBS on Issue Size

Outcome	Sample	Method	RD Estimate	SE	P-Value
Issue Amount (log)	Full Sample	Conventional	-0.004	0.063	0.951
		Bias-Corrected	-0.001	0.063	0.987
		Robust	-0.001	0.064	0.988

Note: RDiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel, and MSE-optimal bandwidth. Includes issuer, industry, country, and daily fixed effects, and controls for liquidity (Bid-Ask spread), with cluster-robust standard errors at the ISIN level.

Additionally, I conduct an independent sample t-test to determine whether the difference in means between the average bond issuance between the two-month periods before (January–February 2023) and after (March–April 2024) is statistically significant and I fail to reject the null hypothesis ($p > 0.05$). The descriptive statistics of the sample appear in Table A.3

Table 3: Summary Statistics for Both Periods

Period	Mean	Standard Deviation (SD)	Sample Size (N)
January–February	528,431,016.7	858,501,009.5	180
March–April	539,623,167.8	1,604,427,811	94

Effects of Private Regulatory Standards

To support my main hypothesis—that public standards have a much larger effect on bond pricing than private-sector ones, I present an alternative analysis testing the impact of a private standard on bond yields. Currently, there are two main private standards: the International Capital Market Association's (ICMA) Green Bond Principles and the Climate Bonds Initiative's (CBI) Climate Bonds Standard and Certification Scheme. ICMA's Green Bond Principles came into effect in April 2014, while CBI's Standard was introduced in December 2019. Due to the limited development of the green bond market in 2014, data availability makes it infeasible to analyze the effect of the Green Bond Principles on bond pricing.

As such, I estimate the effect of the CBI Standard using equation (1) from the main text, applied to a sample of all euro-denominated green bonds traded in the European Union between December 11, 2018, and March 11, 2020, using the announcement date (December 11, 2019) as the cutoff point. As shown by the results, the effect is not only statistically insignificant but also considerably smaller than the effects observed following the introduction of the EU GBS.

Table 4: RD Estimates of CBI Standard

Outcome	Sample	Method	RD Estimate	SE	P-Value
		Conventional	0.009	0.219	0.966
Bond Yields	Full Sample	Bias-Corrected	0.008	0.219	0.972
		Robust	0.008	0.191	0.967

Note: RDiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel, and Mean Squared Error optimal bandwidth with cluster-robust standard errors at the ISIN level.

Appendix B. Robustness Tests

Placebo Cut Points

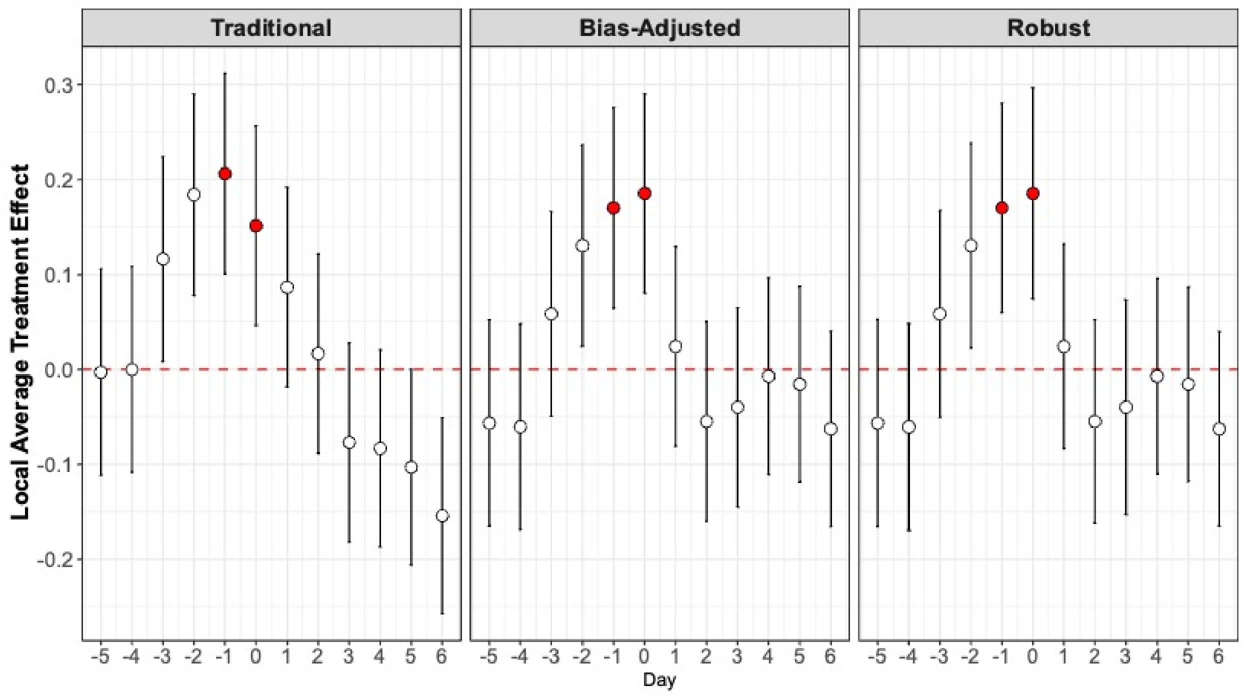

Figure 2: LATE Using different cutdates

Alternative Polynomial Specifications

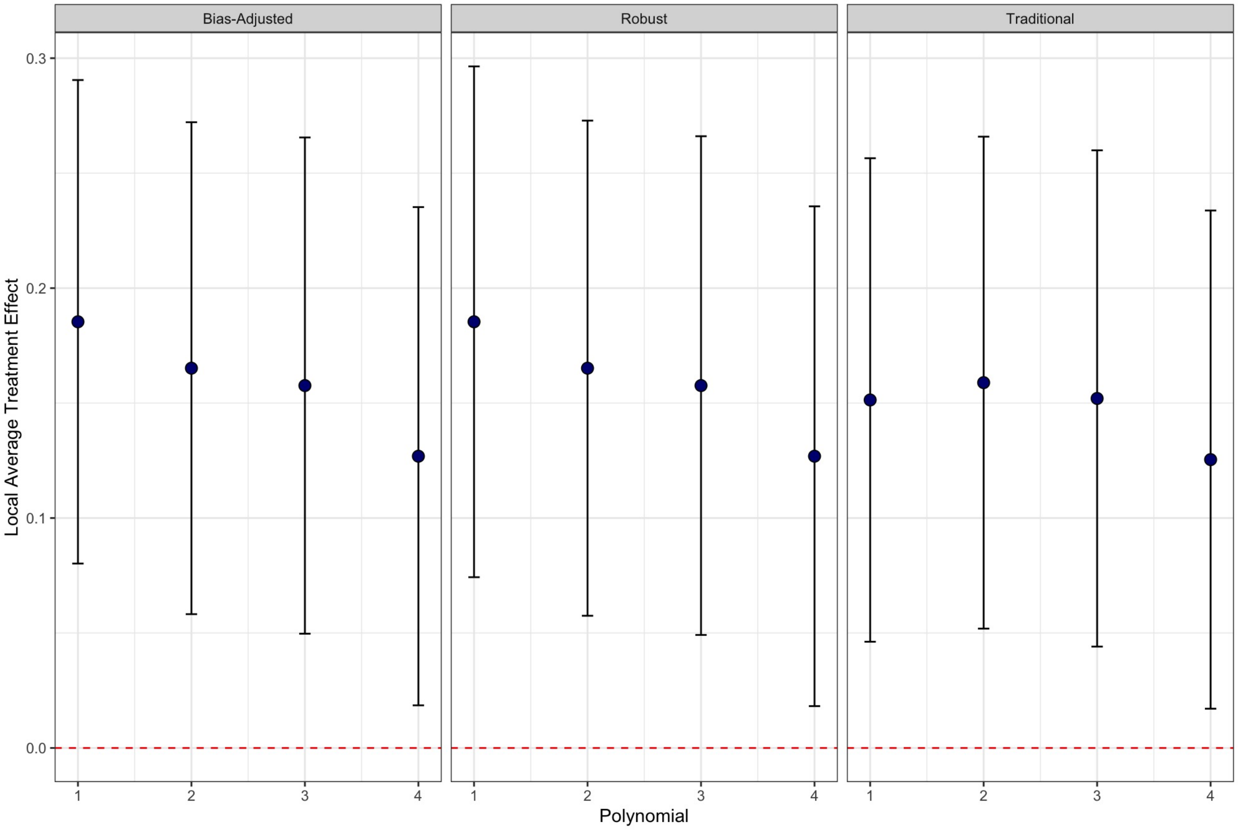

Figure 3: LATE Using different polynomials

Table 5: RD Estimates with different polynomials

Outcome	Polynomial		RD Estimate	SE	P-Value
Bond Yields	1	Conventional	0.151	0.054	0.005
		Bias-Corrected	0.185	0.054	0.001
		Robust	0.185	0.057	0.001
	2	Conventional	0.159	0.055	0.004
		Bias-Corrected	0.165	0.055	0.002
		Robust	0.165	0.055	0.003
	3	Conventional	0.152	0.055	0.006
		Bias-Corrected	0.158	0.055	0.004
		Robust	0.158	0.055	0.004
	4	Conventional	0.125	0.055	0.023
		Bias-Corrected	0.127	0.055	0.022
		Robust	0.127	0.055	0.022

Note: RdiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel and Mean Squared Error optimal bandwidth. Using issuer, industry, country and daily fixed effects and controlling for liquidity (Bid-Ask spread) with cluster-robust standard errors at the ISIN level.

LATE under different Bandwidths

In the main text we estimate the discontinuity within a narrow bandwidth of 33 days in the pre-period and 33 days in the post-period as it is the optimal bandwidth as per CCT's non-parametric approach (Calonico, Cattaneo, and Titiunik 2014). Here we present alternative specifications using varying bandwidths ranging from [-10, 10] days in the pre and post-period to [-40, 40] days as is common in RDiT designs (Hausman and Rapson, 2018).

However, we need to acknowledge that a challenge in bandwidth selection in RDiT designs arises from the fundamental differences between time-based and cross-sectional designs. Given that RDiT relies on variation in the time dimension we cannot grow the sample size while simultaneously shrinking the bandwidth (the proximity to the threshold) as such reducing the bandwidth too much may introduce additional bias due to the time-series properties of the data generating process and reduced number of observations (Hausman and Rapson, 2018).

Figure B.4 shows that the local average treatment effect varies with different bandwidths. We can see that we find both substantive and statistically significant effects of the EU GBS similar to those reported in the main analysis.

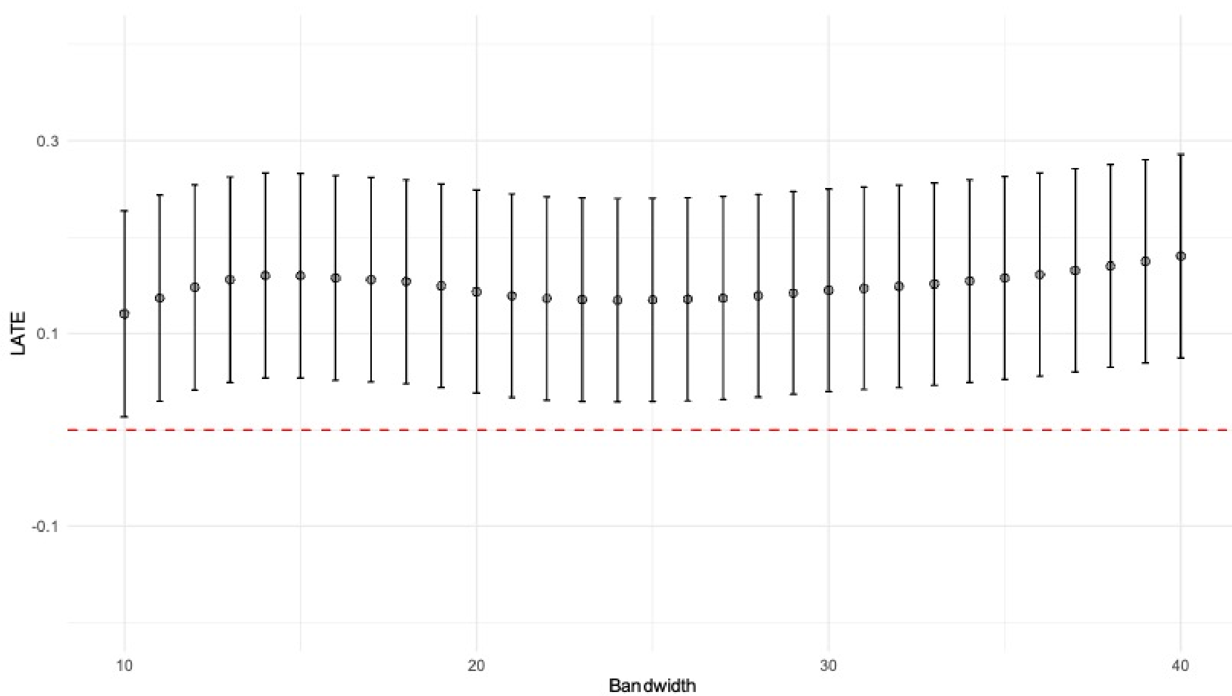

Figure 4: LATE under different bandwidths

Full Figures of Leaders and Laggards Estimates

Figures B.5 and B.6 show the full estimates that appear in Table 2 in the main text.

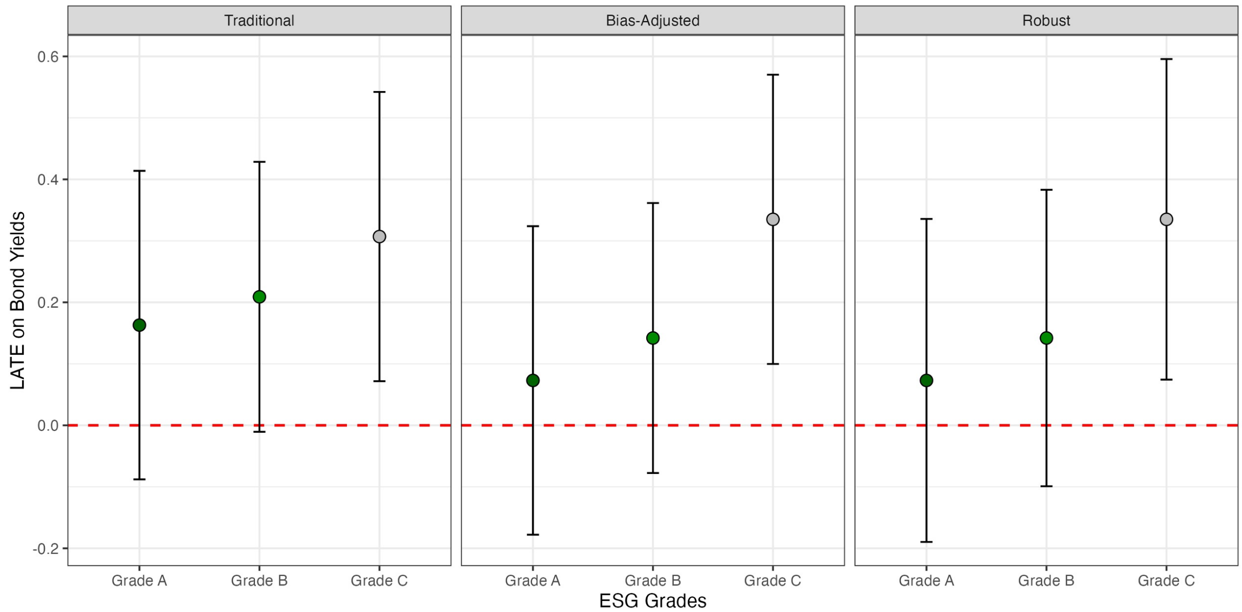

Figure 5: Leaders and Laggards:ESG Scores

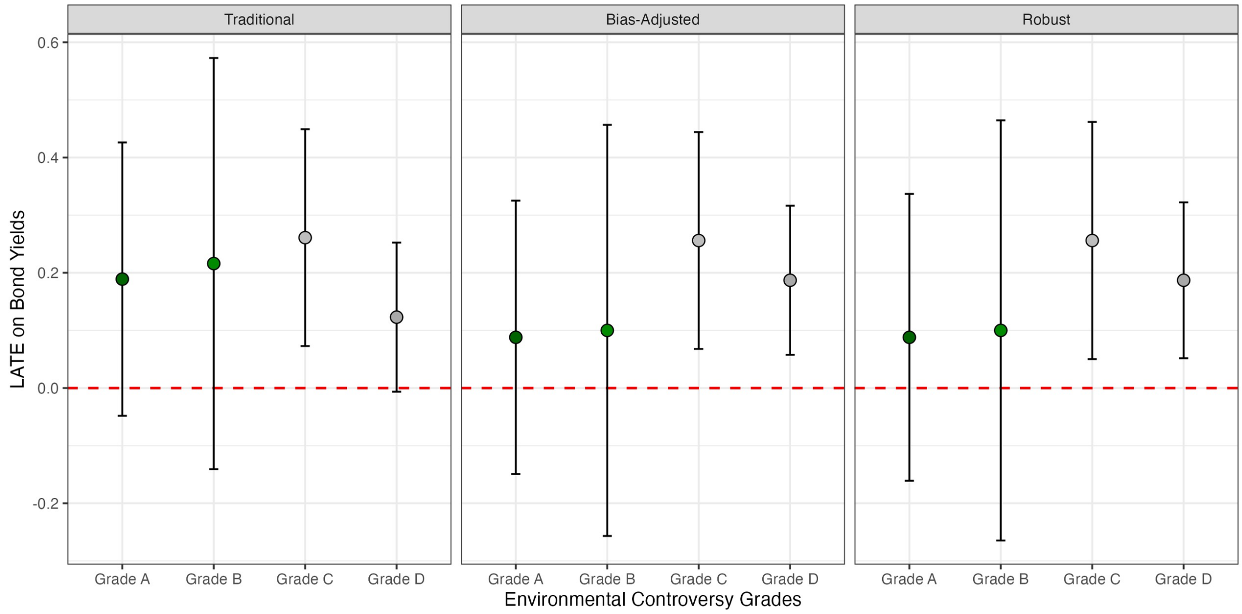

Figure 6: Leaders and Laggards: Environmental Controversies

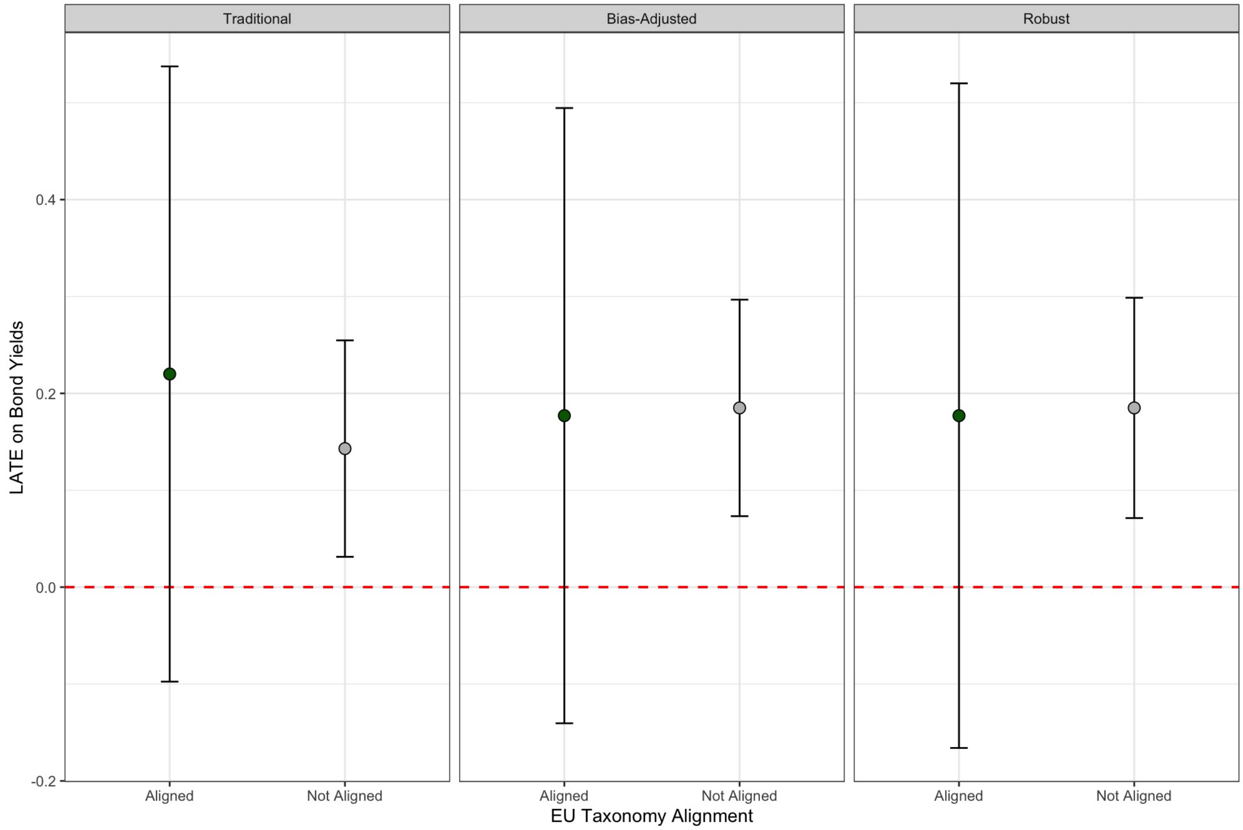

Figure 7: Leaders and Laggards: Taxonomy Alignment

Appendix C. Alternative Specification

Here I present an alternative specification of the leaders and laggards firms to test the robustness of the hypothesis. As mentioned in the main text, Refinitiv grades issuers based on a letter scale that ranges from A+ to D- on issues such as firm's ESG score and its ESG controversies score. In the main analysis, I pool all the bonds across these grades and then condition the effect on the grade. Here I take an alternative approach. Given that the "A" score reflects an excellent relative environmental performance and high degree of transparency in reporting environmental data publicly, issuers graded either A+, A, and A- will serve as a proxy for environmental leaders. Conversely, the C and D score ranges indicate, respectively, satisfactory and poor performance and transparency. Thus, I use issuers with these grades as a proxy for environmental laggards.

As a final measure, Refinitiv tracks whether the framework under which a green bond is issued is compliant with the EU Taxonomy and Green Bond Standard. It is important to note that this measure does not indicate formal certification under the EU Green Bond Standard. Rather, it reflects Refinitiv's assessment that the core components of the bond's issuance framework align with the EU framework. Accordingly, bonds aligned with the taxonomy serve as a proxy for environmental leaders, while those that do not are used as a proxy for laggards.

For brevity, I present only the plots for the robust estimators from the two analyses in Figure C.8. Full regression estimates are reported in Table C.4. As shown, this alternative approach yields qualitatively similar results to those in the main analysis.

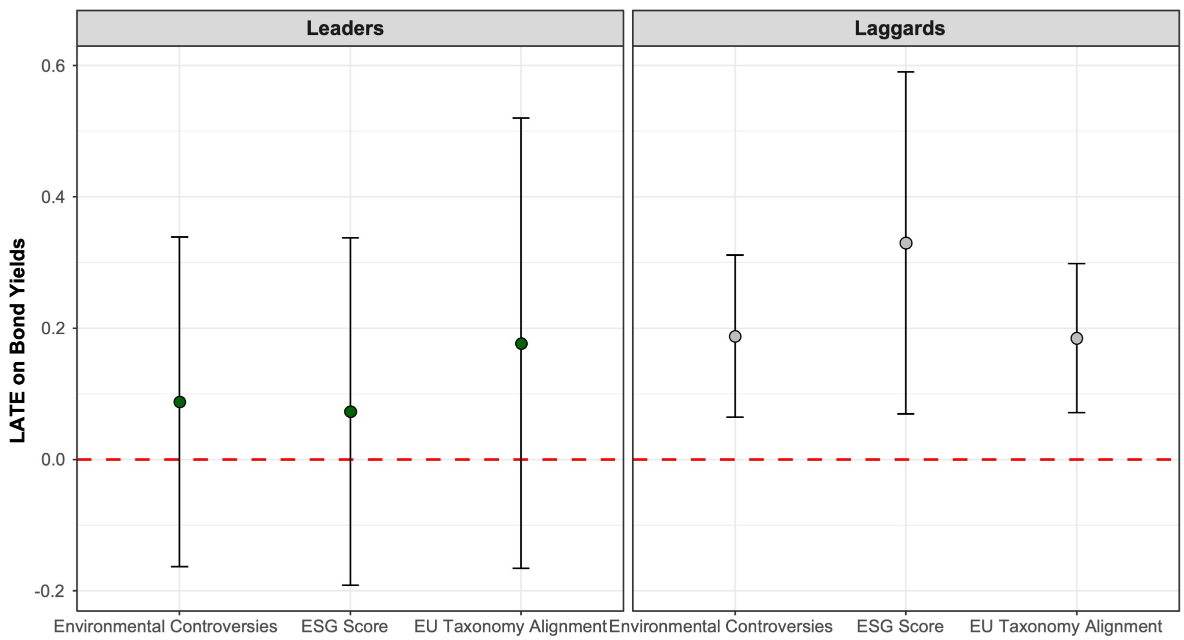

Figure 8: LATE on Bond Yields for leaders and laggards

Table 6: RD Estimates for Environmental Leaders and Laggards

Outcome	Sample	Method	RD Estimate	SE	P-Value
Bond Yields	Controversies Leaders	Conventional	0.189	0.121	0.118
		Bias-Corrected	0.088	0.121	0.465
		Robust	0.088	0.128	0.489
	Controversies Laggards	Conventional	0.137	0.061	0.026
		Bias-Corrected	0.188	0.061	0.002
		Robust	0.188	0.063	0.003
	ESG Leaders	Conventional	0.161	0.128	0.208
		Bias-Corrected	0.073	0.128	0.570
		Robust	0.073	0.135	0.589
	ESG Laggards	Conventional	0.303	0.119	0.011
		Bias-Corrected	0.330	0.119	0.006
		Robust	0.330	0.133	0.013
	EU Aligned	Conventional	0.220	0.162	0.174
		Bias-Corrected	0.177	0.162	0.273
		Robust	0.177	0.175	0.311
	Non-EU Aligned	Conventional	0.143	0.057	0.011
		Bias-Corrected	0.185	0.057	0.001
		Robust	0.185	0.058	0.002

*Note:*RdiT estimates, standard errors, and p-values of the Local Average Treatment Effect on Daily Returns. Estimates with 1st order polynomials, triangular kernel and Mean Squared Error optimal bandwidth. Using issuer, industry, country and daily fixed effects and controlling for liquidity (Bid-Ask spread) with cluster-robust standard errors at the ISIN level.